

# Mohamed Elagamy

AI Engineer | Machine Learning Engineer | Data Scientist

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## PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in production-ready AI systems. Built end-to-end RAG pipelines, NLP transformers, and real-time CV solutions. Using PyTorch, TensorFlow, FastAPI, and Docker.

Passionate about deploying scalable AI that solves real business problems

## TECHNICAL SKILLS

**Languages & Tools:** Python, REST APIs, Docker, Git, Postman

**ML / DL Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn

**NLP:** Text Classification, NER, Tokenization, Embeddings, Semantic Search, Seq2Seq Models

**LLMs & RAG:** RAG Pipelines, Prompt Engineering, LLMs, Fine-Tuning, LoRA, QLoRA, LangChain, Ollama

**Computer Vision:** Image Classification, Object Detection, Object Tracking, Image Segmentation

OCR, Optical Flow, OpenCV, Label Studio

**Backend & APIs:** FastAPI, API Design, Celery

**Databases:** PostgreSQL, MongoDB, TimescaleDB, SQLAlchemy, Alembic, ChromaDB, FAISS, Qdrant

**Streaming & DevOps:** Apache Kafka, Docker Compose, Model Deployment, MLOps, Real-Time Systems

**Languages:** Arabic (Native), English (Professional)

## PROJECTS

### Construction Equipment Utilization Tracker | Real-Time CV Pipeline

*A production-grade, microservices-based computer vision system monitoring construction equipment in real time, streaming utilization analytics to a live dashboard.*

- Fine-tuned YOLOv12n for multi-class equipment detection (excavators, dump trucks) achieving reliable real-time inference
- Implemented multi-object tracking using DeepSORT to maintain persistent Track IDs across video frames
- Engineered region-based Optical Flow to detect articulated machine part motion (e.g., excavator arm vs. stationary tracks)
- Built a state classifier (ACTIVE / INACTIVE) and activity classifier (Digging, Swinging, Loading, Dumping, Waiting) per tracked machine
- Streamed structured JSON telemetry via Apache Kafka → ingested into TimescaleDB for time-series analytics
- Computed per-machine utilization rate (Total Active Time / Total Tracked Time) and served results via a real-time Gradio dashboard

*Tech Stack:* PyTorch · YOLO · DeepSORT · Kafka · TimescaleDB · Docker · Gradio · Python

### Retrieval-Augmented Generation (RAG) System | Document Q&A Platform

*End-to-end RAG system enabling accurate, context-grounded document Q&A using both cloud and local LLMs.*

- Designed and implemented document ingestion, chunking, preprocessing, and embedding pipelines via FastAPI
- Integrated PgVector + PostgreSQL for fast semantic similarity search across large document corpora
- Built an LLM abstraction layer supporting both cloud APIs and local models via Ollama, enabling flexible deployment
- Containerized the entire system with Docker; used Celery for async background task processing

*Tech Stack:* FastAPI · PgVector · PostgreSQL · Docker · Celery · Ollama · LangChain · Python · mongodb

## Arabic Tashkeel Transformer | NLP / Seq2Seq Model

2024

*Character-level Seq2Seq Transformer for Arabic diacritization (Tashkeel), trained and evaluated on Kaggle datasets.*

- Designed a custom character-level tokenizer and preprocessing pipeline tailored for Arabic script
- Implemented a modular Transformer architecture from scratch in PyTorch with training scripts, evaluation notebooks, and checkpoint management
- Exposed inference via a FastAPI REST endpoint, tested with Postman for API validation

*Tech Stack:* [PyTorch](#) · [FastAPI](#) · [Python](#) · [Kaggle](#) · [Postman](#)

## Food Recommendation System | Graduation Project

*AI-powered social-media-driven food recommendation application, integrating NLP classification and a fine-tuned GPT-2 chatbot.*

- Built a Twitter/X API data pipeline to extract and process user tweets for real-time personalized preference analysis
- Trained a two-stage NLP classifier: binary food-content detector followed by multi-class food category classifier
- Fine-tuned a GPT-2-based conversational chatbot for interactive food, ingredient, and recipe recommendations
- Applied feature engineering, text preprocessing, and hyperparameter tuning to maximize classification accuracy

*Tech Stack:* [PyTorch](#) · [Hugging Face](#) · [GPT-2](#) · [Twitter API](#) · [Python](#) · [Scikit-learn](#)

## YOLOv1 Paper Replication | Deep Learning Research

*From-scratch PyTorch replication of the original 2015 YOLOv1 object detection paper for deep architectural understanding.*

- Rebuilt the full YOLOv1 architecture manually in PyTorch, including the custom multi-part loss function from scratch
- Ran controlled experiments on a simplified dataset to validate correctness of implementation against the paper

*Tech Stack:* [PyTorch](#) · [Python](#)

## EDUCATION

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### B.Sc. in Computer Science and Artificial Intelligence | Fayoum University

2020 – 2024

- Faculty of Computers and Artificial Intelligence